



CL1002-Programming Fundamentals– FALL 2023

LAB 14

Pointers



Learning Outcomes

In this lab you are expected to learn the following:

- Pointers



POINTERS

A pointer is a variable that stores an address of a memory location.

Pointer Declaration and initialization

Pointers is declared as follow.

Type* <variable Name>;

```
#include<iostream>
using namespace std;

int main(){

    int *x=NULL;
    // or
    int* x=NULL;
    //or
    int * x=nullptr;

    return 0;
}
```

Address-of operator (&)

The ampersand, **&**, called the **address of operator**, is a unary operator that returns the address of its operand.



```
#include<iostream>
using namespace std;

int main(){

    //declare and initialize string* variable
    string *str=nullptr;
    //declare and initialize string variable
    string temp="hello";
    //store the address of temp to str, both will
    //start pointing to same memory location
    str=&temp;

    //print the address of temp
    cout<<"temp Address "<<&temp<<endl;

    //print value in str
    cout<<"str Value "<<str<<endl;

    return 0;
}
```

```
temp Address 0x7ffdc2d7c440
str Value 0x7ffdc2d7c440
```

Dereference operator (*)

The unary operator * is the dereferencing operator that returns the value of its operand.

```
#include<iostream>
using namespace std;

int main(){

    //declare and initialize string* variable
    string *str=nullptr;
    //declare and initialize string variable
    string temp="hello";
    //store the address of temp to str, both will
    //start pointing to same memory location
    str=&temp;

    //print the address of temp
    cout<<"temp Value "<< temp<<endl;

    //Dereferencing to print value in str
    cout<<"str Value "<< *str<<endl;

    return 0;
}
```

```
temp Value hello
str Value hello
```



Array manipulation using pointer

The identifier of the array is a constant pointer. The elements of the array can be accessed using dereferencing in the following manner.

```
#include<iostream>
using namespace std;

int main(){

    int arr[3]={2,4,8};
    int *p=arr;

    cout<<arr[1]<<endl;
    cout<< *( p+1 )<<endl;

    return 0;
}
```

4
4

Note: You are supposed to pass the test cases for all functions.

Submission Instructions:

1. Create a single cpp file containing all the functions of the problems and main function.
2. Save the **cpp** file with the roll no and task number
e.g. i230001_Q1.cpp
3. Now create a new folder with *ROLLNO_LAB14_SEC* e.g. **i23XXXX_LAB14_A**
4. You need to display your roll no and name before the output of each question.
5. Now you have to submit this zipped file on Google Classroom.
6. If you don't follow the above-mentioned submission instructions, you will be marked **zero**.
7. Plagiarism in the Lab Task will result in **zero** marks in the whole category.



Lab Tasks

Problem 1

A square matrix in which all the entries below the main diagonal are zero is called an upper triangular matrix. Write a function `IsMatrixUpperTriangular` which takes a matrix and returns true if the matrix is upper triangular and false otherwise.

Function Prototype: `IsMatrixUpperTriangular( int[][COLS] , int , int );`

For example, matrix A shown below is upper triangular while matrix B is not upper triangular.

-1	2	7	0
0	5	0	-1
0	0	7	0
0	0	0	0

Matrix A

-1	2	7	0
0	5	0	-1
-9	0	7	0
0	0	0	0

Matrix B

Problem 2

Write a program to take input from user in a 2D Integer array of 5x5. Pass this array to function `csort` which perform the following task.

2 3 1 5 0	0 1 1 0 0
7 1 5 3 1	2 1 5 1 1
2 5 7 8 1	2 3 5 3 1
0 1 5 0 1	3 4 7 5 1
3 4 9 1 5	7 5 9 8 5

The output will be

Note: Sort the Element in Column Wise Manner



Problem 3

Write a Function in C++ to swap the values of two integers using pass by reference.

Example:

In main function:

```
a=10;
```

```
b=20;
```

```
swap(a,b); // function call
```

```
a=20; // now a has the value of b
```

```
b=10; // b has the vale of a
```

Problem 4

Write a program that does the following temperature conversion from one unit to another unit using references. All the functions must take only one parameter and must be declared as void.

From	To	Formula
C	K	$K = (Centigrade + 273.15)$
K	F	$F = Kelvin - 273.1595 + 32$
F	C	$C = (Fahrenheit - 32) \times 59$

Output

```
Enter temperature in Centigrade: 37
Enter temperature in Fahrenheit: 102
Enter temperature in Kelvin: 0

Centigrade to Kelvin: 310.15
Kelvin to Fahrenheit: -276.07
Fahrenheit to Centigrade: -17.7778
```



Problem 5

Write a C++ program to swap the values of 2 integers using their pointers.

```
Enter the first integer: 5
Enter the second integer: 10

Before swapping:
First integer: 5
Second integer: 10

After swapping:
First integer: 10
Second integer: 5
```

Problem 6

Write a program that finds the sum of all the elements in an array. The array should be filled by the user given values using the subscript notation. The variable sum should also be pointer type.